

Inverse-designed metasurfaces for multifunctional spatial frequency filtering

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With the rapid rise in demand for edge computing, there is a need for low-power, compact approaches to information processing. Optical metasurfaces have emerged as a powerful platform for implementing convolution operations in the analog domain that would traditionally be done digitally. However, there has been limited work exploring the full extent to which metasurfaces can be designed to filter the spatial frequency content of incident fields. We experimentally demonstrate inverse-designed metasurfaces that perform spatial frequency filtering by precisely controlling their angular, polarization, and spectral scattering characteristics. To demonstrate the flexibility of our design approach, we show that distinct transfer functions can be implemented for orthogonal polarizations, leading to directional edge detection and blurring. Furthermore, we design metasurfaces with polarization-independent and spectrally multiplexed transfer functions. Multifunctional metasurfaces that encode several filtering functions have the potential to enable new applications in robotics, metrology, and sensing, particularly in resource-constrained scenarios. © 2025 Optica Publishing Group under the terms of the [Optica Open Access Publishing Agreement](#)

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1. INTRODUCTION

Performing image processing operations with minimal latency and power consumption is important for applications such as autonomous vehicles, augmented reality, and large-scale computational imaging. These operations are typically done digitally once an image has been captured, which limits processing speed and discards the phase of the optical field. The computational burden of large two-dimensional convolutions can be reduced by shifting these operations to the optical domain. This is well-established in the context of Fourier optics, where a mask is inserted in the Fourier plane to filter spatial frequencies [1]. Recently, optical metasurfaces have been used for analog image processing via photonic crystal (PhC) resonances [2–8], topological properties of light [9,10], plasmonic resonances [11], and optimized thin-film stacks [12], among others [13–18]. More generally, optical computing as a whole has garnered attention with integrated and free-space approaches achieving remarkable results [19,20]. Metasurface-based processing can enable compact form factors since the dispersion of the resonances carries the desired spatial frequency filtering while foregoing a $4f$ system and Fourier plane mask. Furthermore, metasurfaces can modify the phase, polarization, and amplitude of light, in contrast to the digital case, where only an intensity image is captured.

Photonic inverse design has emerged over the past decade as a promising approach to design a wide variety of optical devices [21–25]. Specifically, topology optimization expands the metasurface design space by framing the design process as

an optimization problem where the metasurface's permittivity is iteratively adjusted to extremize a figure of merit (FoM) that encodes the desired functionality [26]. Several applications of topology-optimized free-space devices have been demonstrated, including high-efficiency gratings [27,28], metalenses [29–32], holograms [33,34], active metasurfaces [35], color filter arrays [36–38], and end-to-end design of optical systems [39–44]. In the context of spatial frequency filtering, it has been computationally demonstrated that inverse design can be used to realize complex transfer functions within periodic structures composed of up to 30 patterned layers [16]. However, the fabrication challenges involved with patterning and aligning tens of layers would likely prevent these structures from being realized outside of advanced semiconductor foundries. More recently, a single-layer inverse-designed metasurface was demonstrated for edge detection, albeit with limited multifunctionality, a small numerical aperture (NA), and poor isotropy [45]. Edge detection aside, there has been limited work considering low-pass filtering or blurring. Examples include collecting reflected light from a metasurface designed for edge detection in transmission to achieve the complementary transfer function [2] and geometric phase metasurfaces limited to operate with circular polarization [46].

In this paper, inverse-designed metasurfaces capable of high-efficiency edge detection and blurring are experimentally demonstrated in the mid-infrared (mid-IR), as schematically depicted in Fig. 1. In contrast to previous work, we leverage inverse design to simultaneously optimize transmission for multiple

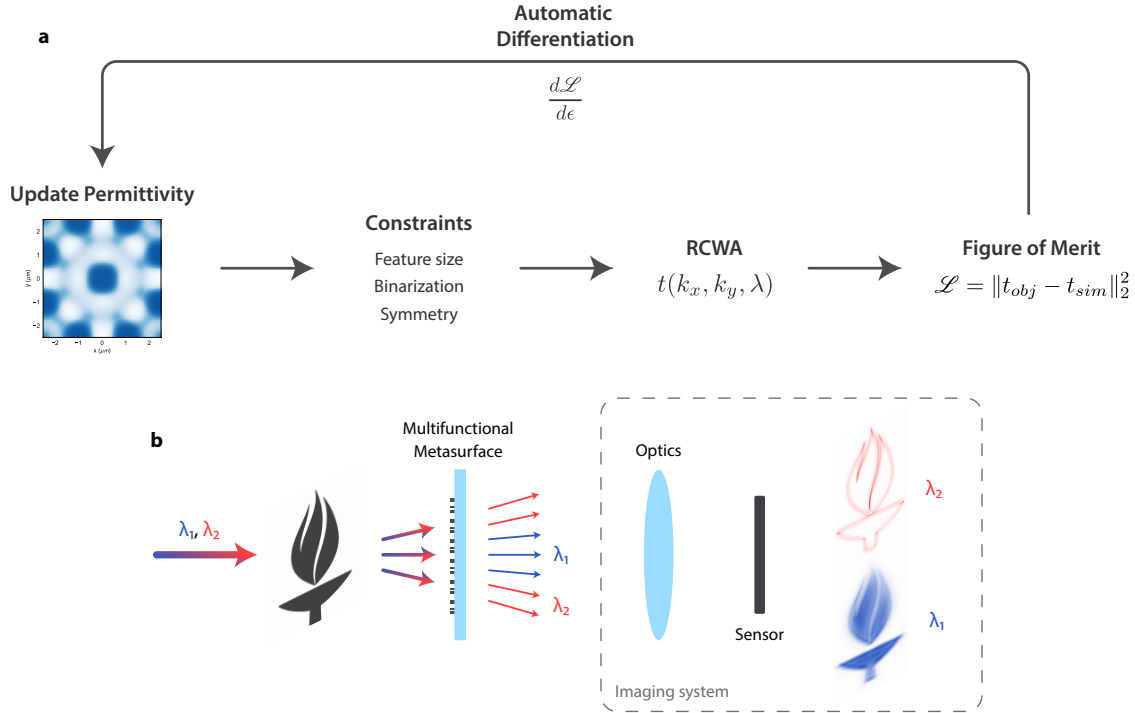


Fig. 1. (a) Depiction of the topology optimization process. The grayscale permittivity represents the structure of a periodic metasurface's unit cell partway through the optimization. (b) Imaging with a multifunctional metasurface imparting distinct transfer functions for λ_1 and λ_2 .

wavelengths, angles of incidence, and polarizations, augmenting the performance compared to rationally designed metasurfaces while restricting ourselves to a single patterned layer for straightforward fabrication. Specifically, we show devices that selectively apply processing operations on s - or p -polarized light to produce filtering whose orientation depends on the polarization of the incident light. Then, we design a metasurface with nearly identical responses for s and p components such that the filtering is polarization independent. Finally, a multifunctional metasurface is demonstrated in which distinct filtering operations are realized on separate spectral bands.

2. METASURFACE DESIGN

A. Metasurfaces as Spatial Frequency Filters

In a traditional $4f$ system, filtering can be realized by placing a mask in the Fourier plane to modify the spatial frequency content of the incident optical field. However, we can simplify this configuration by designing a metasurface whose scattering properties have an angular dependence to modulate the corresponding angular spectrum components. Inserting such a metasurface into the optical system imparts an identical transformation to that of a $4f$ system. Consider the angular spectrum representation of the electric field over a plane in two dimensions for a single polarization [1]:

$$E(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \hat{E}(k_x, k_y) \exp(i(k_x x + k_y y)) dk_x dk_y, \quad (1)$$

where $\hat{E}(k_x, k_y)$ is the Fourier transform of $E(x, y)$, and $k_{x,y}$ are the in-plane k -vectors. In the case of a one-dimensional Laplacian operator ($\partial^2/\partial x^2$), its effect is to add a factor of $-k_x^2$ in the integrand, implying that the transmission coefficient or amplitude transfer function (ATF) must be of the form $|t(k_x, k_y)| \propto k_x^2$.

More generally, we can write down a matrix-valued transfer function H , whose elements are the ATFs corresponding to transformations applied to co- and cross-polarized rays impinging on the metasurface:

$$\begin{bmatrix} E_{p,\text{out}}(k_x, k_y) \\ E_{s,\text{out}}(k_x, k_y) \end{bmatrix} = \underbrace{\begin{bmatrix} t_{pp}(k_x, k_y) & t_{ps}(k_x, k_y) \\ t_{sp}(k_x, k_y) & t_{ss}(k_x, k_y) \end{bmatrix}}_H \begin{bmatrix} E_{p,\text{in}}(k_x, k_y) \\ E_{s,\text{in}}(k_x, k_y) \end{bmatrix}, \quad (2)$$

where $t_{ij}(k_x, k_y)$ denotes the transmission of the polarization state $|j\rangle$ into $|i\rangle$. We aim to design metasurfaces that encode isotropic ATFs to modify the spatial frequency content of the fields incident upon them. The final image thus depends on the combination of the object's polarization state and the ATFs associated with each matrix element in Eq. (2). [Supplement 1](#) provides a more detailed description of the image formation model and examples of simulated images. We will demonstrate how topology optimization can be used to target a variety of transfer functions with different polarization behaviors that enable blurring, edge detection, and multifunctional processing.

B. Inverse Design

Topology optimization has emerged as a powerful technique for designing a variety of optical devices. It provides a flexible platform for realizing functionality that is difficult to achieve with rational design approaches based on simple scatterer geometries. To design metasurfaces for spatial frequency filtering, we seek to minimize the figure of merit (FoM) $\mathcal{L}$:

$$\mathcal{L} = \frac{1}{N} \sum_{ijkl} w(\lambda_i, \theta_j, \phi_k, p_l) |t_{\text{target}}(\lambda_i, \theta_j, \phi_k, p_l) - t_{\text{device}}(\lambda_i, \theta_j, \phi_k, p_l)|^2, \quad (3)$$

where t_{target} and t_{device} represent the desired and simulated transmission coefficients, respectively, for a set of wavelengths $\{\lambda_i\}$, polar angles $\{\theta_j\}$, azimuthal angles $\{\phi_k\}$, and polarization states $\{p_l\}$. N is the total number of these parameter combinations included in the optimization, and w serves as a fixed hyperparameter specifying the relative weight of each combination. In all optimizations, more weight is given to transmission at the smallest and largest polar angles to increase the contrast between the angular stop and pass bands, as well as to improve the overall transmission efficiency. A full breakdown of the optimization parameters can be found in Table S1. The design process is summarized in Fig. 1(a). To minimize the FoM, we use the method of moving asymptotes (MMA) implemented in the open-source NLOpt package and obtain the gradient of the FoM via automatic differentiation [47,48]. To ensure uniform filtering across the device's aperture, we limit ourselves to periodic metasurfaces and use differentiable rigorous coupled-wave analysis (RCWA) to simulate the metasurfaces, allowing gradients to be computed without an explicit adjoint simulation [26,49]. The unit cell is discretized into a 100×100 grid, and at each iteration, its permittivity is mapped to a density variable $\rho \in [0, 1]$, followed by the application of a Gaussian blur and a sigmoidal projection filter to control the minimum feature size and ensure binarization, respectively [16,36,50]. The projection filter takes the following form:

$$\rho(x, y) = \frac{\tanh(\beta\eta) + \tanh(\beta(\rho(x, y) - \eta))}{\tanh(\beta\eta) + \tanh(\beta(1 - \eta))}, \quad (4)$$

where β controls the degree of binarization, and η tunes the erosion or dilation of the pattern. The width of the Gaussian blur kernel is kept fixed, while the sigmoid filter becomes sharper throughout the optimization (by gradually increasing the parameter β) to binarize the structure. Furthermore, the unit cell is forced to have C_4 , σ_x , and σ_y symmetries to increase the isotropy of the ATFs and reduce the number of simulations required. With this symmetry, we only need to simulate azimuthal angles in the range $[0^\circ, 45^\circ]$, and the transmission coefficients of s - and p -rays are restricted to be identical at normal incidence. All of these constraints are differentiable, which makes it straightforward to obtain gradients of the FoM with respect to the unit cell's permittivity. More generally, other symmetries can be imposed on the unit cell's structure, including the absence of any symmetry whereby the unit cell is fully freeform. This could be useful for implementing dichroic responses and polarization conversion, for example.

3. RESULTS

A. Polarization-Selective Filtering

We first aim to design metasurfaces that have different transfer functions for s - and p -polarizations. Edge detection is targeted only for p -polarized rays with no cross-polarized components. The transfer function takes the form:

$$H = \begin{bmatrix} t_{pp}(k_x, k_y) & 0 \\ 0 & 0 \end{bmatrix} \quad (5)$$

indicating that p -polarized rays are transmitted according to $t_{pp}(k_x, k_y)$, while all other polarizations are reflected. Similar transfer functions have been demonstrated in [5–7], where it was shown that, when illuminating with linearly polarized light, the edge orientation is determined by the orientation of the incident

polarization. We choose p -polarized light arbitrarily, but the same procedure can be applied to perform the filtering operation only on s -polarized light, and we include such a design in Fig. S7 in Supplement 1. Furthermore, our design approach can be extended to target not only sp states, but also circularly polarized and HV states oriented in the lab frame. The symmetry of the unit cell forces $t_{ss} = t_{pp}$ at normal incidence, but by lifting the symmetry, it is possible to design a device with distinct transmission coefficients at normal incidence for orthogonal polarizations.

The metasurface's lattice constant is $5 \mu\text{m}$, slightly larger than the target wavelength of $\lambda_0 = 4.65 \mu\text{m}$. While it is possible for diffraction orders to propagate, we find that they have minimal transmission within the NA of our imaging system based on RCWA calculations (Figs. S8–S11 in Supplement 1). It is also possible to explicitly suppress transmission into higher diffraction orders during the optimization as in [16], but we find that this is not strictly necessary in practice. The metasurface consists of a $1 \mu\text{m}$ (or $\sim \lambda_0/5$) layer of crystalline silicon (c-Si) on a $500 \mu\text{m}$ sapphire substrate (MTI Inc). The thickness has been chosen based on commercial availability and empirical experimentation to produce good optimization results while remaining thin enough to allow plasma etching of the smallest features. We assume the refractive index of silicon and sapphire to be 3.48 and 1.65, respectively [51,52]. The unit cell permittivity is optimized to produce the following transmission amplitude target for p -polarized light:

$$|t_{pp}(\theta, \phi)| = 1 - \exp(-\theta^2/\sigma_\theta^2), \quad (6)$$

while minimizing transmission for all angles of s -polarized light. In Eq. (6), σ_θ sets the width of the angular rejection band, and the transformation $[k_x, k_y] = k_0 \sin(\theta)[\cos(\phi), \sin(\phi)]$ has been applied to express the optimization problem in terms of polar and azimuthal angles. We choose this form of the transfer function to realize a general high-pass filter, not necessarily the Laplacian operator that was described above. It is possible to modify the optimization target to be a Laplacian if desired, such that all light within the NA is filtered accordingly. Equation (6) also naturally lends itself to enforcing high transmission at angles beyond σ_θ to maximize the edge signal. A larger σ_θ will transmit higher spatial frequency components to produce an image with finer edge features, albeit with lower intensity. With this in mind, we choose σ_θ to be relatively small to produce prominent edge features. Optimization details for all devices presented in this work can be found in Supplement 1.

The optimized unit cell permittivity and a scanning electron microscope (SEM) image of the fabricated structure are shown in Figs. 2(a) and 2(b). In Fig. 2(c), we show the simulated transmission coefficient amplitude obtained via RCWA for p -polarized light with a free-space wavelength of $\lambda_0 = 4.5 \mu\text{m}$ up to a NA of 0.65. Evidently, $|t_{pp}|$ is highly isotropic and agrees well with the targeted transmission function. At normal incidence, the transmission amplitude is 0.03, and it peaks at 0.88 near 15° . A key feature of our design is that the average transmission amplitude for p -rays is approximately 0.8 for $0.2 \leq k_{x,y} \leq 0.6$, which means that high spatial frequencies are transmitted efficiently and contribute to a stronger edge signal. Furthermore, $|t_{ss}|$ remains low for all $k_{x,y}$ points (see Fig. S12 in Supplement 1), ensuring significantly weaker transmission of the s -polarized signal. Indeed, the average transmitted power for s -rays within an NA of 0.65 is 16%.

To measure the transmission coefficient, we weakly focus polarized, collimated light onto the device with a $f = 150 \text{ mm}$

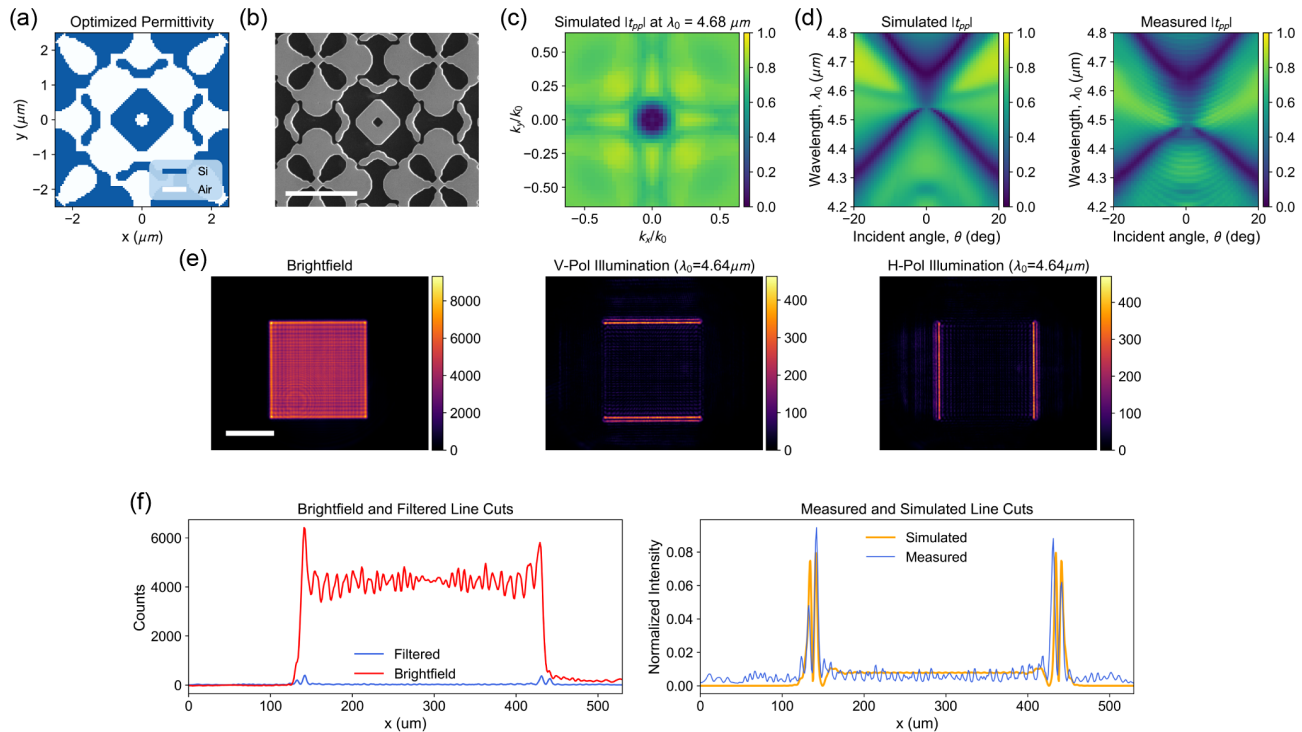


Fig. 2. Polarization-selective high-pass filtering. (a) Optimized metasurface unit cell permittivity, where blue regions are silicon and white regions are air; (b) SEM micrograph of the fabricated metasurface (scale bar 3 μm); (c) simulated ATF for p -polarized light; (d) simulated and measured p -polarized transmission coefficient amplitudes for a set of wavelengths and incident angles; (e) measured brightfield image with no metasurface and filtered images with linear vertical and horizontal incident polarizations in the middle and right panels, respectively (scale bar 130 μm); (f) line cut comparison of the brightfield and filtered (V -pol) images as well as a comparison of cuts through the measured and simulated images normalized to the intensity of the brightfield square's center.

Si lens. The light transmitted through the device is collected by a $f = 100$ mm Si lens and directed to a thermopile photodetector (Thorlabs). The sample is mounted on an automated rotation stage to set the incident angle while a tunable mid-IR quantum cascade laser (MIRcat, Daylight Solutions) scans the incident wavelength. Absorption in the sapphire substrate is accounted for by taking reference measurements through it at all (θ, λ) pairs for each polarization. A diagram of the measurement setup is included in Fig. S4(a). Figure 2(d) shows the simulated and experimentally measured transmission coefficient amplitudes along the $\phi = 0^\circ$ azimuth as a function of incident polar angle and wavelength. The measured and simulated results are in agreement, aside from an overall approximate 2% blue shift of the measured spectra, attributed to pattern erosion during the fabrication process. In Fig. S5 in Supplement 1, we have included an additional simulation and measurement of angle-resolved $|t_{pp}|$ along the $\phi = 45^\circ$ azimuthal direction for further confirmation that the metasurface is working as intended. The s -polarized transmission coefficient amplitude is also measured and shown in Fig. S12, confirming that this polarization component is mostly rejected. In the measured spectra, there is a weak ripple due to leaky Fabry–Perot modes of the sapphire substrate. Its amplitude is reduced by adding an SiO_2 anti-reflective coating to the sapphire's backside following Si patterning (see Section 5). The refractive index of SiO_2 at $\lambda_0 = 4.5 \mu\text{m}$ is around 1.36 [53].

We further verify the functionality of by inserting the metasurface into an imaging system, shown in Fig. S4(b) and image a 260 μm -wide square onto an InSb sensor (FLIR A6700) with a $f = 12.7$ mm ZnSe aspheric lens. In order to ensure that all

scattered rays are filtered correctly, we fabricate the metasurface over a 1 mm^2 area and place it within 500 μm of the imaging target. As discussed in [18], the distance between the target and the metasurface does not affect the filtering operation, but as the distance increases, the metasurface's area must scale such that all scattered rays within the NA go through the device. In the left panel of Fig. 2(e), a brightfield image with no device present is shown captured with an incident wavelength of $\lambda_0 = 4.64 \mu\text{m}$. With the device in place between the imaging target and lens, we observe sharp vertical and horizontal edges when illuminating with H - and V -polarized light, respectively. This behavior is expected because the s -rays are rejected by the metasurface for all angles, while only p -rays scattered away from normal incidence are transmitted. In Fig. 2(f), we take line cuts through the vertically polarized and brightfield panels of Fig. 2(e) showing the relative intensity of the filtered image. Although the edges seem faint in comparison to the brightfield image, this is the expected behavior and is demonstrated by comparing the filtered line cut to one from a simulated image using the simulated ATFs of the metasurface [Fig. 2(f) right panel; simulated image shown in Fig. S1]. To compare the simulated and experimentally measured line cuts directly, we normalize the measured image to the average intensity in the center of the brightfield image scaled by the incident power of the laser. The simulated image is normalized to an input power of unity. There is good agreement between the simulated and measured images, which quantitatively confirms that the metasurface is operating near its theoretical efficiency. Additional noise in the measured images stems from stray light in the optical system due to reflections from the custom imaging target made of chromium (Cr). We compare

the efficiency of our devices to that of the polarization-selective metasurface presented in [7] in terms of the η_{avg} metric described therein, which is the edge intensity normalized to the input image intensity. This is a useful metric because it accounts for the overall edge transmission efficiency agnostic to the exact form of the ATF. The peak intensity of the line cuts in Fig. 2(f) corresponds to η_{avg} for our metasurface and is near 7% (both in simulation and experiment), while in [7], it is less than 2%. Our device achieves such high efficiency due to its narrow angular stop band and large transmission up to an NA of at least 0.65, enabling more spatial frequency content to be collected by the imaging system.

As a complement to edge detection, we designed a polarization-selective low-pass, or blurring, filter. In this case, higher spatial frequency components of the p -polarized field are rejected, while the s -polarized field is fully transmitted for all angles of incidence, such that the transfer function is

$$H = \begin{bmatrix} t_{pp}(k_x, k_y) & 0 \\ 0 & 1 \end{bmatrix}. \quad (7)$$

This produces a blurring effect along the direction of the incident linear polarization's orientation. The target transmission is a Gaussian function defined by

$$|t_{pp}(\theta, \phi)| = \exp(-\theta^2/\sigma_\theta^2), \quad (8)$$

with $\sigma_\theta = 7.5^\circ$. In Eq. (8), σ_θ represents the width of the angular pass band, such that reducing σ_θ leads to a stronger blurring effect. As with our definition of the transmission target for the high-pass filter, Eq. (8) is suitable to realize devices that retain their target transmission at large angles of incidence. Compared to the work presented in [2] where the low-pass functionality was achieved in reflection, we design our metasurface to operate in transmission to enable easier integration into optical systems. Gaussian filters can apodize coherent imaging systems to reduce ringing artifacts introduced by the sharp spatial frequency cutoff at the imaging system's NA [54]. It has also been shown that apodization can reduce the sensitivity of coherent imaging systems to aberrations [55]. Owing to the ability of metasurfaces to manipulate the phase and amplitude of light, it may be possible for a metasurface to simultaneously apodize and correct aberrations in an optical system [56].

The device is designed to operate near $\lambda_0 = 4.5 \mu\text{m}$. By updating the target transmission function, the optimization procedure generates the unit cell permittivity shown in Fig. 3(a), and an SEM image of the fabricated device is shown in Fig. 3(b). The simulated transmission amplitude for s - and p -polarized light is shown in Fig. 3(c) as a function of k_x and k_y at $4.56 \mu\text{m}$, demonstrating an isotropic low-pass filtering behavior for p -rays. A comparison to the target transmission amplitude is presented in Fig. S14 and shows good agreement. The simulated transmission coefficients are also shown in Fig. 3(d) across wavelength and incident angle. We confirm that the fabricated device operates as expected by measuring its transmission coefficients $|t_{pp}|$ and $|t_{ss}|$, as shown in Fig. 3(e). Measured and simulated transmission spectra along the $\phi = 45^\circ$ azimuth are included in Supplement 1. Again, we find very good agreement between the simulated and experimentally measured transmission spectra, indicating high-fidelity fabrication. To further characterize the low-pass functionality, we capture images of a square imaging target [Fig. 3(f)] when illuminated with V - and H -polarized light in the left and right panels, respectively, with a wavelength of $4.56 \mu\text{m}$. With V -polarized illumination, the vertical edges are preserved while the horizontal edges are

strongly blurred, a consequence of the ATF's polarization selectivity. Conversely, with H -polarized light, the blurring direction is horizontal. More generally, the blurring direction can be tuned continuously by varying the orientation of the incident linear polarization.

We quantify the polarization-dependent blurring by experimentally measuring the contrast from a series of line arrays. Specifically, we fabricated a custom imaging target with sets of line arrays of various periods. Images of horizontal lines are captured with and without the metasurface for orthogonal linear polarizations. The contrast, defined as $C = (I_{\text{max}} - I_{\text{min}})/(I_{\text{max}} + I_{\text{min}})$, is computed from background-subtracted images and plotted in Fig. 3(g) as a function of line array spatial frequency (normalized to the free-space wavevector's magnitude). With V -polarized light, the line contrast is comparable to that of the H -polarized and brightfield cases for small spatial frequencies but drops off dramatically for line arrays with smaller pitch. The contrast trend qualitatively agrees with the simulated ATFs and quantifies the effect of the metasurface within the imaging system. In theory, these curves should match a cut through the simulated ATFs, but in practice, they differ for a few reasons. First, there are aberrations in our optical system from using a single aspheric lens for imaging. Second, stray light propagates because of reflections from the Cr imaging target and from the metasurface. Coherent illumination leads to interference effects that magnify the detrimental effects of stray light, such as ghost images. Finally, the lines consist of sharp edges, so the measured contrast does not only correspond to a single spatial frequency. These additional components add coherently and produce artifacts that cause deviations from the simulated ATFs. To excite only a single spatial frequency, a continuously varying amplitude grating would be necessary, which is difficult to fabricate. Despite these factors, the curves in Fig. 3(g) clearly demonstrate the directional contrast reduction introduced by the metasurface.

B. Polarization-Independent Filtering

So far, we have considered polarization-selective devices that filter s - or p -polarized light; yet, in many cases, polarization-independent filtering is desired, such that $t_{ss}(k_x, k_y) = t_{pp}(k_x, k_y)$. It is difficult to achieve this performance with rationally designed metasurfaces because it generally requires precise dispersion engineering [6]. To demonstrate this behavior, we optimize a metasurface such that it encodes high-pass filtering for s - and p -polarizations using the same target function as the device in Fig. 2. The transfer function in this case is

$$H = \begin{bmatrix} t_{pp}(k_x, k_y) & 0 \\ 0 & t_{ss}(k_x, k_y) \end{bmatrix}. \quad (9)$$

As before, the target ATF is defined by Eq. (6), and we optimize over several hundred iterations to minimize Eq. (3). We use the same period of $5 \mu\text{m}$ and Si thickness of $1 \mu\text{m}$ as for the devices of the previous section, targeting a wavelength of $\lambda_0 = 4.5 \mu\text{m}$. The resulting unit cell permittivity and corresponding SEM of the fabricated device are shown in Figs. 4(a) and 4(b), respectively. The simulated ATFs for p - and s -polarization both exhibit high-pass behavior, in contrast to the polarization-selective devices [Fig. 4(c)]. The high spatial frequency content is transmitted efficiently, and the p -polarized transfer function is almost perfectly isotropic. For the s -polarization, there is some variation in the

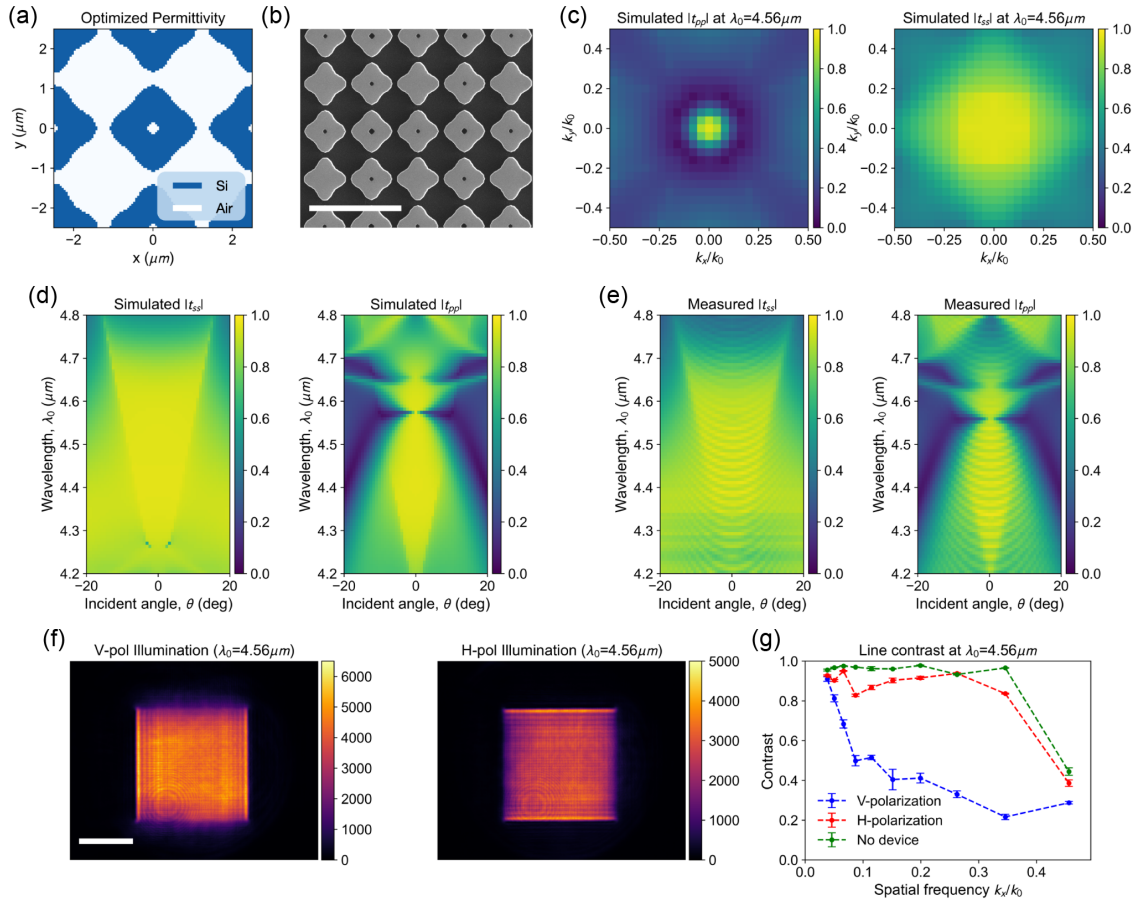


Fig. 3. Polarization-selective low-pass filtering. (a) Final optimized unit cell permittivity for the polarization-selective low-pass filter; (b) SEM image of the fabricated metasurface (scale bar 5 μm); (c) simulated co-polarized transmission coefficient magnitudes for s - and p -polarization, which represents the ATF of the metasurface; (d) simulated and (e) measured transmission coefficient magnitudes for s - and p -polarization across wavelength and incident angle along the $\phi = 0$ azimuth; (f) measured images of a square object filtered through the metasurface with V - and H -polarized illumination (scale bar 130 μm); (g) measured contrast for a series of line arrays, as a function of normalized spatial frequency, with no metasurface (green), metasurface with V -pol (blue), and metasurface with H -pol (red). The error bars correspond to the standard deviation of contrast calculated from 10 slices through the images of the line arrays.

azimuthal direction, meaning that the edge contrast along certain azimuths will be slightly reduced. In both cases, the transmission efficiency remains high up to a large NA. Simulated and experimentally measured angle-resolved transmission spectra are in excellent agreement for both polarizations [Figs. 4(d) and 4(e)].

Images of a Caltech logo captured at $\lambda_0 = 4.55 \mu\text{m}$ (slightly red-shifted compared to the design wavelength due to fabrication errors) are shown in Fig. 4(f) and confirm that the metasurface extracts high-contrast edge information for vertically and horizontally polarized illumination. These results highlight that, despite azimuthal variation in the s -polarized ATF, the image is not noticeably affected, an effect that has been observed elsewhere [5,6]. Brighter spots appear in the measured images where two edges meet, which is expected based on image simulations presented in Fig. S2 in Supplement 1. Horizontal line cuts through the images in Fig. 4(g) show the similarity for both polarization states. Assuming no cross-polarized contribution, we expect the efficiency to be similar to that of the polarization-selective device because s - and p -rays contribute to orthogonal edges rather than doubling the efficiency of any individual edge. Indeed, in Fig. S3, we include simulated images of a square object along with a line cut through it, demonstrating efficient transmission of edges

with comparable intensity to those from Fig. 2. Compared to the polarization-independent metasurface in [6] with simulated edge efficiencies below 3% for vertical edges, our device transmits edges with an efficiency of above 10%. Our inverse design approach enables the discovery of structures with a narrow angular pass band and high efficiency transmission, such that strong edge features are produced.

C. Multifunctional Filtering

We have demonstrated filtering functionality whose polarization response can be readily altered by reformulating the desired transfer function elements embedded within Eq. (3). Now, we explore the ability to extend the functionality of these devices by targeting unique, isotropic ATFs for different wavelengths of light. Near $\lambda_0 = 4.65 \mu\text{m}$, the metasurface performs edge detection for p -rays while suppressing transmission of s -rays. Conversely, near $\lambda_0 = 4.35 \mu\text{m}$, a low-pass ATF is targeted for p -rays with s -rays fully transmitted. Essentially, we combine the two functionalities of the devices shown in Figs. 2 and 3 into a single metasurface targeting those ATFs at distinct wavelengths. Specifically, the transfer function now takes the form:

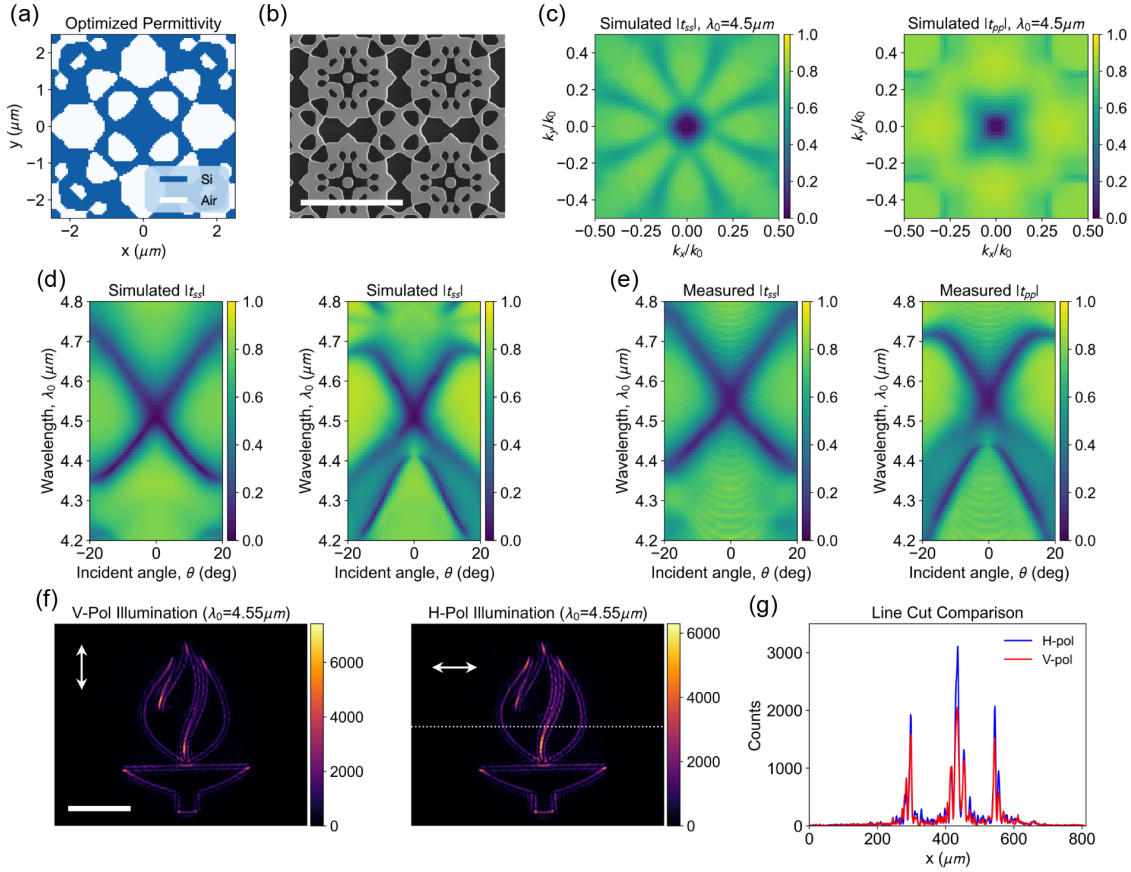


Fig. 4. Polarization-independent high-pass filtering. (a) Optimized unit cell permittivity; (b) SEM of the fabricated metasurface (scale bar 5 μm); (c) simulated and (d) measured transmission coefficient amplitudes for s - and p -polarized light across wavelength and incident angle; (e) simulated ATFs for s - and p -polarized light through the metasurface; (f) measured images of a Caltech logo through the metasurface with V - and H -polarized light (scale bar 200 μm); (g) horizontal line cuts through the two images in (f).

$$H = \begin{bmatrix} t_{pp}(k_x, k_y, \lambda) & 0 \\ 0 & t_{ss}(k_x, k_y, \lambda) \end{bmatrix}, \quad (10)$$

where the wavelength dependence is now explicit. We optimize six wavelengths simultaneously: three centered around 4.375 μm following Eq. (8) and three centered around 4.675 μm following Eq. (6). The optimized unit cell and an SEM image of the fabricated device are shown in Figs. 5(a) and 5(b), and RCWA calculations of the transmission amplitudes are shown in Fig. 5(c). The measured transmission of the fabricated metasurface is shown in Fig. 5(d) and agrees closely with the simulations. The optimization algorithm converges on a structure hosting two resonances that cross near the Γ -point near 4.5 μm to achieve the desired spectrally multiplexed transfer functions. Simulations also show that the ATFs at these wavelengths are almost perfectly isotropic, meaning that the filtering action is virtually identical for all azimuths [Figs. 5(e) and 5(h)].

With 4.37 μm light, we observe a polarization-selective blur, as shown in Fig. 5(f), which shows images of a Caltech logo. With H -polarized light, the contrast is reduced in the horizontal direction, as demonstrated in the line cuts presented in Fig. 5(g). Illuminating with orthogonal polarization decreases the contrast along the vertical direction instead. Thus, in Fig. 5(g), the contrast for the V -polarization and the brightfield case are similar. The p -polarized pass band of the optimized metasurface is wider than was desired, leading to a weaker blurring effect, yet it still approximates

a Gaussian function and attenuates higher spatial frequencies. Because of this, we use a smaller object for a clearer demonstration of the blurring effect. The measured contrast curve is shown in Fig. S13, showing that the contrast is reduced for higher spatial frequencies for V -polarized light, as expected. In Fig. 5(i), images of the Caltech logo are captured with both H - and V -polarized illumination at $\lambda_0 = 4.7 \mu\text{m}$. With V -polarized light, horizontal edges are highlighted, while with H -polarized light, vertical edges are highlighted. In the experiment, we use 4.7 μm light because of the slight red shift of the transmission spectra relative to the RCWA calculations.

4. DISCUSSION AND CONCLUSION

We have demonstrated multifunctional inverse-designed metasurfaces for spatial frequency filtering at mid-IR wavelengths. These devices control transmission for several angles of incidence, wavelengths, and polarizations to realize high-efficiency and isotropic ATFs that preserve their target transmission for large NAs. In our experiments, we inserted the metasurface immediately behind the object for convenience. This configuration limits the filtered field of view to the size of the metasurface and potentially restricts the object to be back-illuminated. However, a principal feature of the metasurfaces shown in this work is their potential to be placed immediately above an image sensor to realize compact form factors [18]. As light passes through the imaging lens, the object is

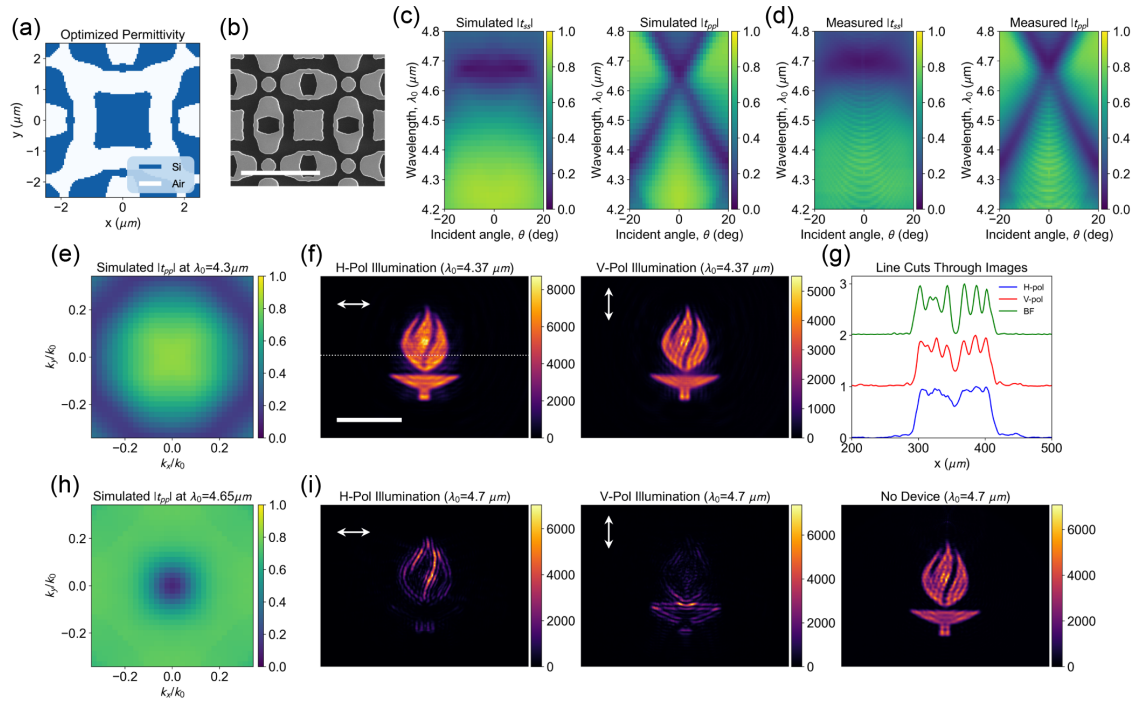


Fig. 5. Spectrally multiplexed filtering. (a) Optimized unit cell permittivity of the multifunctional metasurface; (b) SEM image of the fabricated metasurface (scale bar 5 μm); (c) simulated and (d) measured transmission coefficient amplitude for s - and p -polarized rays; (e) simulated ATF at a wavelength of 4.3 μm ; (f) measured images of resolution target bar groups through the metasurface with a laser wavelength of 4.37 μm for V - and H -polarized light (scale bar 180 μm); (g) line cuts through the vertical bar group from (f) showing the difference in contrast for the two polarization states; (h) simulated ATF for a wavelength of 4.65 μm ; (i) measured images of a Caltech logo for V - and H -polarized light with a wavelength of 4.7 μm [same scale bar as in (f)].

magnified and the spatial frequencies are scaled accordingly. Thus, to design an image-processing metasurface placed directly above an image sensor, the spatial frequencies of the targeted ATF in the optimization must be scaled to be commensurate with those of the magnified image. A potential challenge in this regard is increasing the metasurface aperture to cover the entire image sensor, which can be on the order of 1 cm^2 . In principle, electron beam lithography (EBL) can be used, but the writing times become long and field stitching errors must be mitigated, which limits the practicality of mass production. Alternatively, techniques such as deep UV projection lithography or nano-imprint lithography (NIL) can be utilized, both of which have been used to fabricate large-area metasurfaces [57–59]. To fabricate inverse-designed structures in a commercial semiconductor foundry, additional design rules must be strictly enforced, which is beyond the scope of this work but has been explored by other groups [60,61].

Even though, in principle, the effect of a $4f$ system with any linear Fourier plane mask can be realized with a single metasurface filter to reduce bulkiness, there are practical limitations to this approach. Importantly, the filtering is determined by the dispersion relation of the resonant modes, such that it may be difficult to implement extremely narrow pass or stop bands corresponding to sharp angular responses. Furthermore, because low- Q modes typically have broad angular features, realizing filters with narrow spatial frequency stop or pass bands over a wide spectral range is particularly challenging for metasurfaces and may be better suited to a Fourier plane mask in a $4f$ system. Metasurfaces will excel in situations where wavelength- and polarization-dependent filtering is required. Here, we have shown a multiwavelength metasurface with two targeted ATFs, but it is possible to increase the number of multiplexed ATFs in the optimization. Having said that, different

transfer functions cannot be realized within an arbitrarily narrow spectral band. Exploring the degree to which this is possible may be the subject of future work.

We have found that devices designed to have high-pass ATFs typically retain high transmission to larger incident angles than low-pass ATFs retain their extinction. In other words, we find that it is difficult to realize strong extinction at all angles outside the pass band of the low-pass filter. A possible explanation for this effect is that, in the absence of any resonance, light can be transmitted efficiently through the device as determined by the direct scattering process [62]. For the high-pass filter, this means that the metasurface should only be resonant near normal incidence. Conversely, the low-pass filter requires many resonances that couple strongly to zeroth-order free-space light at all k -points except near normal incidence, which is challenging to achieve in a single-layer metasurface. It is also important to highlight that there is a fundamental trade-off between the spectral operating bandwidth and the total efficiency of transmitted edge information for high-pass devices. As the desired spectral bandwidth increases, there should be a corresponding increase in the angular width of the rejection band since this implies a lower- Q resonance. When the metasurface rejects a larger range of angles (or spatial frequencies), the total transmitted edge signal is reduced, which in turn limits the efficiency and contrast of the edge features.

Targeting the mid-IR spectral band benefits our designs, as the longer wavelength enables inverse-designed structures to have deeply sub-wavelength features that can be readily fabricated via EBL (see Section 5). Furthermore, there are many applications in the mid-IR wavelength band for sensing, imaging, and metrology [63–65]. Accessing this feature size regime is helpful for realizing

high-performance metasurfaces. In principle, however, these structures can be scaled to any target wavelength, and with EBL systems capable of writing features on the order of tens of nanometers, there is potential for demonstrating inverse-designed metasurface filters at near-IR and visible wavelengths. Because the refractive index of silicon is similar in the telecom range as in the mid-IR, our current designs could be directly scaled down to operate at these wavelengths. In the visible range, transparent materials such as titania, gallium nitride, or silicon carbide could be used while accounting for their respective indices of refraction in the optimizations.

The capacity of our platform to realize multifunctional devices may be useful for next-generation imaging systems, where different spatial-frequency bands of the object can be accessed by modulating the wavelength or polarization of light. When combined with spectral color splitters, the decomposed spatial frequencies could be extracted in a single shot [36]. In the context of accelerating machine vision tasks with analog processing, the metasurfaces presented in this work can replace computationally intensive 2D convolutions, and they may become particularly useful in low-power, resource-constrained settings, as discussed in Supplement 1.

Looking forward, we envision multifunctional metasurfaces capable of filtering spatial frequency content not only based on wavelength, but also based on diffraction order, which is used for multi-channel imaging. Furthermore, we have only targeted transfer functions that modify the amplitude of light, but there is an opportunity to realize metasurfaces with tailored angle-dependent phase that can unlock novel functionality such as aberration correction and compression of free-space propagation [66–68]. In the future, multilayer patterned metasurfaces could enable these complex functions, potentially leading to the miniaturization of optical systems and efficient information processing.

5. METHODS

Our devices are fabricated on commercially bought crystalline Si epitaxially grown on a sapphire substrate (MTI Inc). The Si and sapphire thicknesses are approximately 1 μm and 500 μm thick, respectively. The sample is cleaned by sonicating in acetone and rinsing with isopropyl alcohol, followed by oxygen plasma. After baking the sample on a hotplate at 180°C, ZEP520A (Zeon Corporation) is spin-coated on the silicon surface. The sample is then baked again to remove the solvent from the resist. Next, aquaSAVE (Mitsubishi Chemical Group) is spin-coated on top of the resist to act as a conductive charge dissipation layer. The pattern is written with an EBP 5200 (Raith) EBL system. Following exposure, aquaSAVE is removed by immersing the sample in DI water. The resist is developed in ZED developer (Zeon Corporation) and rinsed with MIBK and IPA. Subsequently, a 20 nm thick alumina hard mask is evaporated onto the sample and lifted off with Remover PG (Kayaku). The pattern is transferred to Si with a pseudo-Bosch etch using a mixture of SF_6 and C_4F_8 gases. To remove the hard mask, the sample is immersed in a 1:1 mixture of NH_4OH and H_2O_2 . We add an additional 840 nm layer of SiO_2 deposited via PECVD on the backside of the sample, which acts as an anti-reflection coating.

All optimizations and RCWA calculations were performed on a high-performance computing cluster. At each iteration, the transmission for every combination of wavelength, angle, and polarization is simulated in parallel with eight Intel CPU cores (Skylake, Cascadelake, and Icelake processors).

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

Supplemental document. See Supplement 1 for supporting content.

REFERENCES

1. J. Goodman, *Introduction to Fourier Optics*, 3rd ed. (Roberts & Company, 2005).
2. C. Guo, M. Xiao, M. Minkov, *et al.*, "Isotropic wavevector domain image filters by a photonic crystal slab device," *J. Opt. Soc. Am. A* **35**, 1685–1691 (2018).
3. H. Kwon, A. Cordaro, D. Sounas, *et al.*, "Dual-polarization analog 2D image processing with nonlocal metasurfaces," *ACS Photonics* **7**, 1799–1805 (2020).
4. H. Kwon, D. Sounas, A. Cordaro, *et al.*, "Nonlocal metasurfaces for optical signal processing," *Phys. Rev. Lett.* **121**, 173004 (2018).
5. Y. Zhou, H. Zheng, I. I. Kravchenko, *et al.*, "Flat optics for image differentiation," *Nat. Photonics* **14**, 316–323 (2020).
6. M. Cotrufo, A. Arora, S. Singh, *et al.*, "Dispersion engineered metasurfaces for broadband, high-NA, high-efficiency, dual-polarization analog image processing," *Nat. Commun.* **14**, 7078 (2023).
7. M. Cotrufo, S. Singh, A. Arora, *et al.*, "Polarization imaging and edge detection with image-processing metasurfaces," *Optica* **10**, 1331–1338 (2023).
8. A. Cordaro, H. Kwon, D. Sounas, *et al.*, "High-index dielectric metasurfaces performing mathematical operations," *Nano Lett.* **19**, 8418–8423 (2019).
9. F. Zangeneh-Nejad and R. Fleury, "Topological analog signal processing," *Nat. Commun.* **10**, 2058 (2019).
10. T. Zhu, Y. Lou, Y. Zhou, *et al.*, "Generalized spatial differentiation from the spin Hall effect of light and its application in image processing of edge detection," *Phys. Rev. Appl.* **11**, 034043 (2019).
11. T. Zhu, Y. Zhou, Y. Lou, *et al.*, "Plasmonic computing of spatial differentiation," *Nat. Commun.* **8**, 15391 (2017).
12. W. Xue and O. D. Miller, "High-NA optical edge detection via optimized multilayer films," *J. Opt.* **23**, 125004 (2021).
13. L. Wesemann, E. Panchenko, K. Singh, *et al.*, "Selective near-perfect absorbing mirror as a spatial frequency filter for optical image processing," *APL Photonics* **4**, 100801 (2019).
14. J. Zhou, H. Qian, C.-F. Chen, *et al.*, "Optical edge detection based on high-efficiency dielectric metasurface," *Proc. Natl. Acad. Sci. USA* **116**, 11137–11140 (2019).
15. B. T. Swartz, H. Zheng, G. T. Forcherio, *et al.*, "Broadband and large-aperture metasurface edge encoders for incoherent infrared radiation," *Sci. Adv.* **10**, eadk0024 (2024).
16. H. Wang, W. Jin, C. Guo, *et al.*, "Design of compact meta-crystal slab for general optical convolution," *ACS Photonics* **9**, 1358–1365 (2022).
17. P. Huo, L. Tan, Y. Jin, *et al.*, "Broadband and parallel multiple-order optical spatial differentiation enabled by Bessel vortex modulated metalens," *Nat. Commun.* **15**, 9045 (2024).
18. H. Wang, C. Guo, Z. Zhao, *et al.*, "Compact incoherent image differentiation with nanophotonic structures," *ACS Photonics* **7**, 338–343 (2020).
19. A. Cordaro, B. Edwards, V. Nikkhah, *et al.*, "Solving integral equations in free space with inverse-designed ultrathin optical metagratings," *Nat. Nanotechnol.* **18**, 365–372 (2023).
20. A. Silva, F. Monticone, G. Castaldi, *et al.*, "Performing mathematical operations with metamaterials," *Science* **343**, 160–163 (2014).
21. S. Molesky, Z. Lin, A. Y. Piggott, *et al.*, "Inverse design in nanophotonics," *Nat. Photonics* **12**, 659–670 (2018).

22. J. Jensen and O. Sigmund, "Topology optimization for nano-photonics," *Laser Photonics Rev.* **5**, 308–321 (2011).
23. T. W. Hughes, M. Minkov, I. A. D. Williamson, *et al.*, "Adjoint method and inverse design for nonlinear nanophotonic devices," *ACS Photonics* **5**, 4781–4787 (2018).
24. L. Su, A. Y. Piggott, N. V. Sapra, *et al.*, "Inverse design and demonstration of a compact on-chip narrowband three-channel wavelength demultiplexer," *ACS Photonics* **5**, 301–305 (2018).
25. Z. Li, R. Pestourie, Z. Lin, *et al.*, "Empowering metasurfaces with inverse design: principles and applications," *ACS Photonics* **9**, 2178–2192 (2022).
26. O. D. Miller, "Photonic design: from fundamental solar cell physics to computational inverse design," Ph.D. thesis (University of California, 2012).
27. D. Sell, J. Yang, S. Doshay, *et al.*, "Large-angle, multifunctional meta-gratings based on freeform multimode geometries," *Nano Lett.* **17**, 3752–3757 (2017).
28. S. Ans, F. Zamkotsian, and G. Demésy, "Topology optimization of blazed gratings under conical incidence," *J. Opt. Soc. Am. A* **41**, 1531–1543 (2024).
29. M. Mansouree, H. Kwon, E. Arbabi, *et al.*, "Multifunctional 2.5D metasurfaces enabled by adjoint optimization," *Optica* **7**, 77–84 (2020).
30. Z. Li, R. Pestourie, J.-S. Park, *et al.*, "Inverse design enables large-scale high-performance meta-optics reshaping virtual reality," *Nat. Commun.* **13**, 2409 (2022).
31. E. Bayati, R. Pestourie, S. Colburn, *et al.*, "Inverse designed metalenses with extended depth of focus," *ACS Photonics* **7**, 873–878 (2020).
32. H. Chung, F. Zhang, H. Li, *et al.*, "Inverse design of high-NA metalens for maskless lithography," *Nanophotonics* **12**, 2371–2381 (2023).
33. S. So, J. Kim, T. Badloe, *et al.*, "Multicolor and 3D holography generated by inverse-designed single-cell metasurfaces," *Adv. Mater.* **35**, 2208520 (2023).
34. Y. Yin, Q. Jiang, H. Wang, *et al.*, "Multi-dimensional multiplexed metasurface holography by inverse design," *Adv. Mater.* **36**, 2312303 (2024).
35. H. Chung and O. D. Miller, "Tunable metasurface inverse design for 80% switching efficiencies and 144° angular deflection," *ACS Photonics* **7**, 2236–2243 (2020).
36. G. Roberts, C. Ballew, T. Zheng, *et al.*, "3D-patterned inverse-designed mid-infrared metaoptics," *Nat. Commun.* **14**, 2768 (2023).
37. C. Kim, J. Hong, J. Jang, *et al.*, "Freeform metasurface color router for deep submicron pixel image sensors," *Sci. Adv.* **10**, eadn9000 (2024).
38. P. B. Catrysse, N. Zhao, W. Jin, *et al.*, "Subwavelength Bayer RGB color routers with perfect optical efficiency," *Nanophotonics* **11**, 2381–2387 (2022).
39. Z. Lin, C. Roques-Carmes, R. Pestourie, *et al.*, "End-to-end nanophotonic inverse design for imaging and polarimetry," *Nanophotonics* **10**, 1177–1187 (2021).
40. G. Arya, W. F. Li, C. Roques-Carmes, *et al.*, "End-to-end optimization of metasurfaces for imaging with compressed sensing," *ACS Photonics* **11**, 2077–2087 (2024).
41. K. Wei, X. Li, J. Froech, *et al.*, "Spatially varying nanophotonic neural networks," *Sci. Adv.* **10**, eadp0391 (2024).
42. L. Huang, Q. A. A. Tanguy, J. E. Fröch, *et al.*, "Photonic advantage of optical encoders," *Nanophotonics* **13**, 1191–1196 (2024).
43. H. Zheng, Q. Liu, Y. Zhou, *et al.*, "Meta-optic accelerators for object classifiers," *Sci. Adv.* **8**, eabo6410 (2022).
44. H. Zheng, Q. Liu, I. I. Kravchenko, *et al.*, "Multichannel meta-imagers for accelerating machine vision," *Nat. Nanotechnol.* **19**, 471–478 (2024).
45. Z.-X. Zhou, X.-Y. Zhang, H.-R. Qin, *et al.*, "Inverse design of multiplexable meta-devices for imaging and processing," *ACS Photonics* **12**, 246–252 (2024).
46. W. Fu, D. Zhao, Z. Li, *et al.*, "Ultracompact meta-imagers for arbitrary all-optical convolution," *Light Sci. Appl.* **11**, 62 (2022).
47. S. G. Johnson, "The NLOpt nonlinear-optimization package".
48. K. Svanberg, "A class of globally convergent optimization methods based on conservative convex separable approximations," *SIAM J. Optim.* **12**, 555–573 (2002).
49. W. Jin, W. Li, M. Orenstein, *et al.*, "Inverse design of lightweight broadband reflector for relativistic lightsail propulsion," *ACS Photonics* **7**, 2350–2355 (2020).
50. F. Wang, B. S. Lazarov, and O. Sigmund, "On projection methods, convergence and robust formulations in topology optimization," *Struct. Multidisc. Optim.* **43**, 767–784 (2011).
51. D. Chandler-Horowitz and P. M. Amirtharaj, "High-accuracy, midinfrared ($450\text{ cm}^{-1} \leq \omega \leq 4000\text{ cm}^{-1}$) refractive index values of silicon," *J. Appl. Phys.* **97**, 123526 (2005).
52. I. H. Malitson, F. V. Murphy, and W. S. Rodney, "Refractive index of synthetic sapphire," *J. Opt. Soc. Am.* **48**, 72–73 (1958).
53. I. H. Malitson, "Interspecimen comparison of the refractive index of fused silica," *J. Opt. Soc. Am.* **55**, 1205–1209 (1965).
54. M. E. Krisl, "Aposidation of coherent optical systems," Ph.D. thesis (University of Rochester, 1978).
55. J. P. Mills and B. J. Thompson, "Effect of aberrations and apodization on the performance of coherent optical systems," *J. Opt. Soc. Am. A* **3**, 704–716 (1986).
56. Y. Shao, R. Lupo, J. Jiang, *et al.*, "Multifunctional spaceplates for optical aberration correction," *ACS Photonics* **11**, 1753–1760 (2024).
57. J.-S. Park, S. W. D. Lim, A. Amirzhan, *et al.*, "All-glass 100 mm diameter visible metalens for imaging the cosmos," *ACS Nano* **18**, 3187–3198 (2024).
58. J. Kim, J. Seong, W. Kim, *et al.*, "Scalable manufacturing of high-index atomic layer–polymer hybrid metasurfaces for metaphotonics in the visible," *Nat. Mater.* **22**, 474–481 (2023).
59. V. J. Einck, M. Torfeh, A. McClung, *et al.*, "Scalable nanoimprint lithography process for manufacturing visible metasurfaces composed of high aspect ratio TiO₂ meta-atoms," *ACS Photonics* **8**, 2400–2409 (2021).
60. M. F. Schubert, A. K. C. Cheung, I. A. D. Williamson, *et al.*, "Inverse design of photonic devices with strict foundry fabrication constraints," *ACS Photonics* **9**, 2327–2336 (2022).
61. A. Y. Piggott, E. Y. Ma, L. Su, *et al.*, "Inverse-designed photonics for semiconductor foundries," *ACS Photonics* **7**, 569–575 (2020).
62. S. Fan and J. D. Joannopoulos, "Analysis of guided resonances in photonic crystal slabs," *Phys. Rev. B* **65**, 235112 (2002).
63. A. Titti, A. Leitis, M. Liu, *et al.*, "Imaging-based molecular barcoding with pixelated dielectric metasurfaces," *Science* **360**, 1105–1109 (2018).
64. B. H. Stuart, *Infrared Spectroscopy: Fundamentals and Applications* (Wiley, 2004).
65. L. Dong, X. Yang, C. Zhang, *et al.*, "Nanogapped Au antennas for ultra-sensitive surface-enhanced infrared absorption spectroscopy," *Nano Lett.* **17**, 5768–5774 (2017).
66. K. Shastri and F. Monticone, "Nonlocal flat optics," *Nat. Photonics* **17**, 36–47 (2023).
67. M. Pahlevaninezhad and F. Monticone, "Multi-color spaceplates in the visible," *ACS Nano* **18**, 28585–28595 (2024).
68. K. Shastri, O. Reshef, R. W. Boyd, *et al.*, "To what extent can space be compressed? Bandwidth limits of spaceplates," *Optica* **9**, 738–754 (2022).